Detectable Effect Cluster Analysis: A Novel Machine-Learning Subgroup Analysis Method for Drug Rescue

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Introduction

ALS drug development has been plagued by high clinical trial failure rates. Subgroup analysis is a key tool used to account for patient heterogeneity, but current methods fall short. DEC analysis systematically groups and analyzes patients based on predicted disease path, creating more homogeneous patient subgroups with reduced noise around the endpoint.

Methods

To perform DEC Analysis, a multivariate, non-linear machine-learning model trained using the PRO-ACT dataset to predict disease progression was used to rank-order trial participants by predicted disease progression. An initial 50 subgroups were systematically expanded by adjusting prediction thresholds in 2% nearest-neighbor increments until the full analysis set was reached. To analyze the subgroups, a matrix was plotted in which each block was derived using distinct upper and lower thresholds. A series of analyses were then performed to assess variance (RMSE), treatment effect (TE), effect size, and P value, thus developing a series of heat maps that revealed subgroups with favorable conditions for detecting a significant effect size, whether it be through enhanced TE, lowered variance, or a combination. The method was applied to the Ceftriaxone-ALS (1, NCT00349622) and Topiramate-ALS (2) data sets available from the US NINDS.

Method: Detectable Effect Cluster (DEC) Analysis

- Rank-order a group of trial participants by predicted outcome
- Using the risk-based rank-order, divide trial into 50 subgroups
- Systematically expand each subgroup in 2% nearest neighbor increments
- Perform a statistical analysis on each subgroup; calculate RMSE, treatment effect, effect size and P value
- Plot subgroups on a matrix in which each block represents a subgroup with distinct upper and lower percentile limits

References

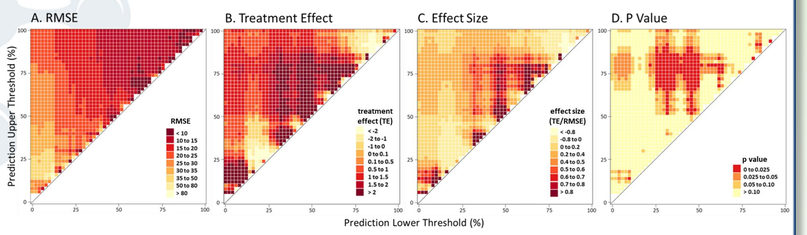
1. Cudkowicz, et al.(2014). Safety and efficacy of ceftriaxone for amyotrophic lateral sclerosis: A multi-stage, randomised, double-blind, placebo-controlled trial. The Lancet Neurology, 13(11), 1083–1091. doi: 10.1016/S1474-4422(14)70222-4
2. Cudkowicz, et al. (2003). A randomized, placebo-controlled trial of topiramate in amyotrophic lateral sclerosis. Neurology, 61(4), 456–464. doi: 10.1212/wnl.61.4.456

Results

To focus on the development of DEC analysis, we used the 285 patients in the Ceftriaxone-ALS dataset who remained on study for one year. The 2:1 allocation of the full study was retained in this group and included 190 treated and 95 placebo patients, which allowed us to perform an exploratory analysis to generate a hypothesis to test in a second analysis. We randomly separated the 190 treated patients into two groups, one for the exploratory analysis and a second to test the hypothesis. One-year predictions using our validated percent expected vital capacity model were made for all patients in the dataset. A broad central region, a "hot spot," where moderately progressing patients localized was detected and a subgroup, representing predicted one-year decline in percent expected vital capacity between 15% and 25% was selected to determine whether it could be detected in the test set. Examination of the test group confirmed the results of the exploratory analysis.

The Topiramate-ALS trial (1) proved useful for developing DEC analysis. In contrast to ceftriaxone, which overall had a slightly positive, non-significant TE, the topiramate trial reported a negative TE for the primary endpoint. DEC analysis using decline in ALSFRS-R as the endpoint was performed and failed to see any positive hot spots. Rather, the TE matrix showed broad zones of negative TEs. This experiment provides a strong negative control for DEC analysis.

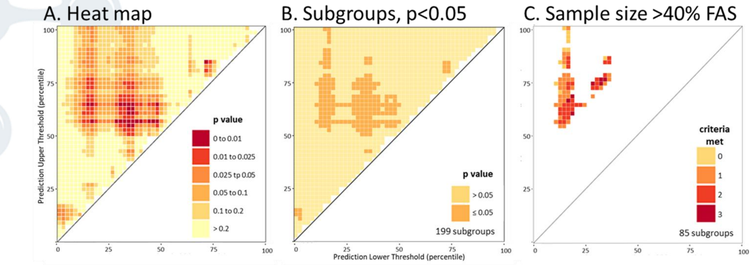
DEC Analysis Simulation



Simulated clinical trial with detectable treatment effect in a subgroup but not the full analysis set. Three hundred patient records were randomly drawn from the PRO-ACT database, then randomly allocated 1:1 to "active" and "placebo" arms and a subtle treatment effect positively correlated to the baseline creatinine value of each patient (R^2 = 0.7) was applied in such a way that the full analysis set would not reach significance for the ALSFRS-R outcome. DEC analysis was performed using the predicted log hazard value for the estimated time to 50% vital capacity to rank order the patients, and the resultant statistics from each subgroup are presented above. The four figures represent visualizations of (A) variance, calculated as Root Mean Square Error (RMSE), (B) treatment effect (in points of the ALSFRS-R scale), (C) effect size (treatment effect ÷ RMSE), and (D) P value. The detection of a subgroup with a statistically significant treatment effect can be seen in panel D within the red cluster in the upper right central portion.

Data used for the development of the models in this study were obtained from the PRO-ACT Database. We thank Drs. Merit Cudkowicz and Jon Glass for suggesting the Ceftriaxone-ALS dataset. This work was partially funded by a grant from the US National Institutes of Health (NIMH), grant R43 MH122925.

Ceftriaxone DEC Analysis: Hypothesis Generation

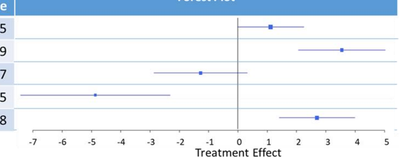


Proof-of-Concept using Ceftriaxone in ALS Clinical Trial Data Set. The 285 patients enrolled in trial NCT00349622 who survived one year from baseline were included. This was a 2:1 treatment to placebo trial. To test DEC analysis, the data were reconfigured as a 1:1 trial. Ninety-five treated patients were set aside for hypothesis testing and 190 (95 treated, 95 placebo) were used for the hypothesis generating. (A) A hot spot identified using predicted change in % expected vital capacity. (B) Subgroups with P>0.05 or P≤0.05 were isolated. (C) Hot spots with P≤ 0.05 representing 40% or more of the trial were identified using three criteria, including P<0.04, simulated power of the subgroup >85, and predicted power >85. One of these, with predicted one-year decline in % expected vital capacity between 15% and 25% was selected for further study.

Ceftriaxone DEC Analysis: Hypothesis Testing

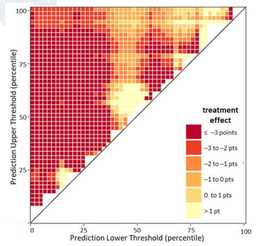
	Ceftriaxone Group	Treatment Effect	RMSE	Effect Size	P value
A.	Full Analysis Set	1.115	7.770	0.144	0.325
B.	Selected subgroup	3.544	7.727	0.459	0.019
C.	High Exclusion subgroup	−1.278	5.109	−0.250	0.427
D.	Low Exclusion subgroup	−4.868	7.770	−0.627	0.065
E.	Test group	2.697	7.705	0.250	0.038

Forest Plot



Proof-of-Concept using Ceftriaxone in ALS Clinical Trial Data Set. Treatment effect (TE), RMSE, effect size and P values were calculated for (A) the ceftriaxone study full analysis set, (B) the subgroup selected above with predicted percent expected vital capacity decline between 15% and 25%, the excluded subgroups (C) above and (D) below the selected subgroup, and (E) the test group that was set aside to test the reproducibility of the selected subgroup. The selected subgroup (B) had a P value of 0.019 with a TE of 3.544 points on the 48-point ALSFRS-R scale over the course of one year. This result was verified in the test group (E) , which had a P value of 0.038 and an average TE of 2.697 points.

Topiramate DEC Analysis



DEC Analysis Negative Control. The 296 patients enrolled in the Topiramate-ALS trial were included. A hot spot was not identified in the topiramate trial. Rather, strong evidence of a negative treatment effect as measured by ALSFRS-R progression was found throughout the trial subgroups.

Conclusions

DEC analysis organizes clinical trial participants in an unbiased way into subgroups that are more homogeneous than the full trial population

- Method reveals "hot spots" of detectable treatment effects that could form the basis for a subsequent successful clinical trial

Numerous ways to implement this approach are envisioned:

- "Rescue" of drugs that failed late-stage clinical development
- Application to all-comers trials with the goal of identifying patients with detectable effects to seamlessly expand into a fully powered trial